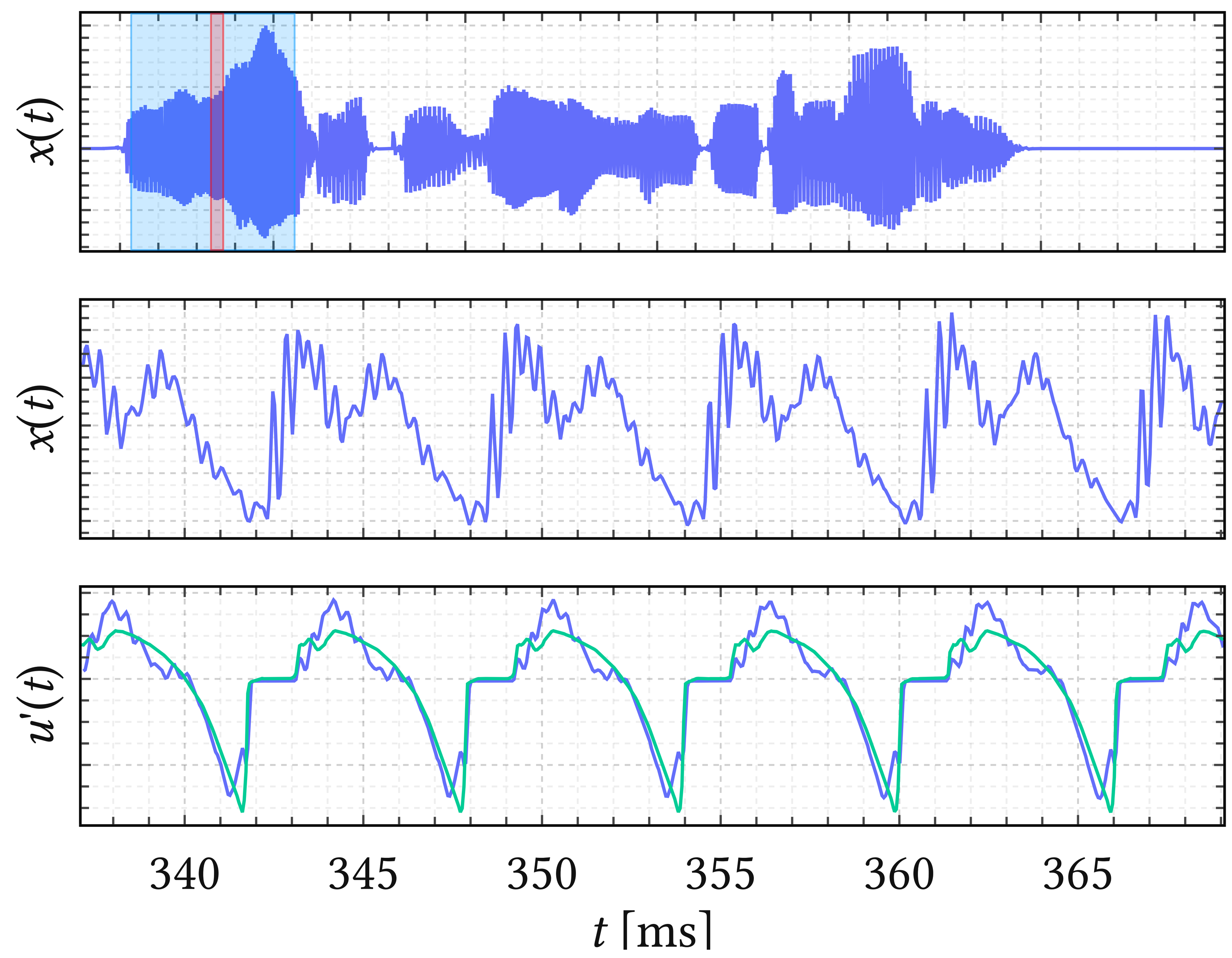
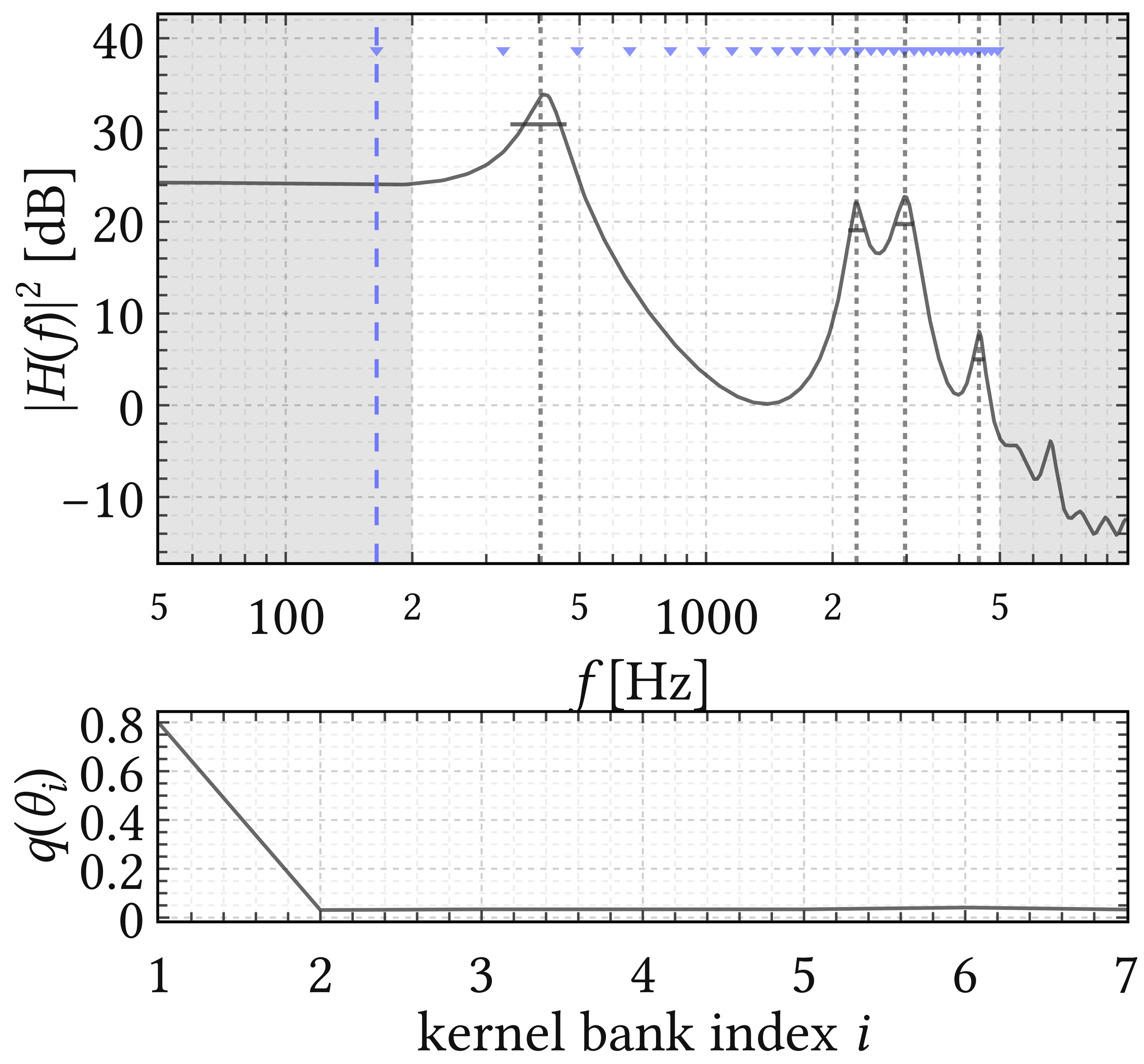
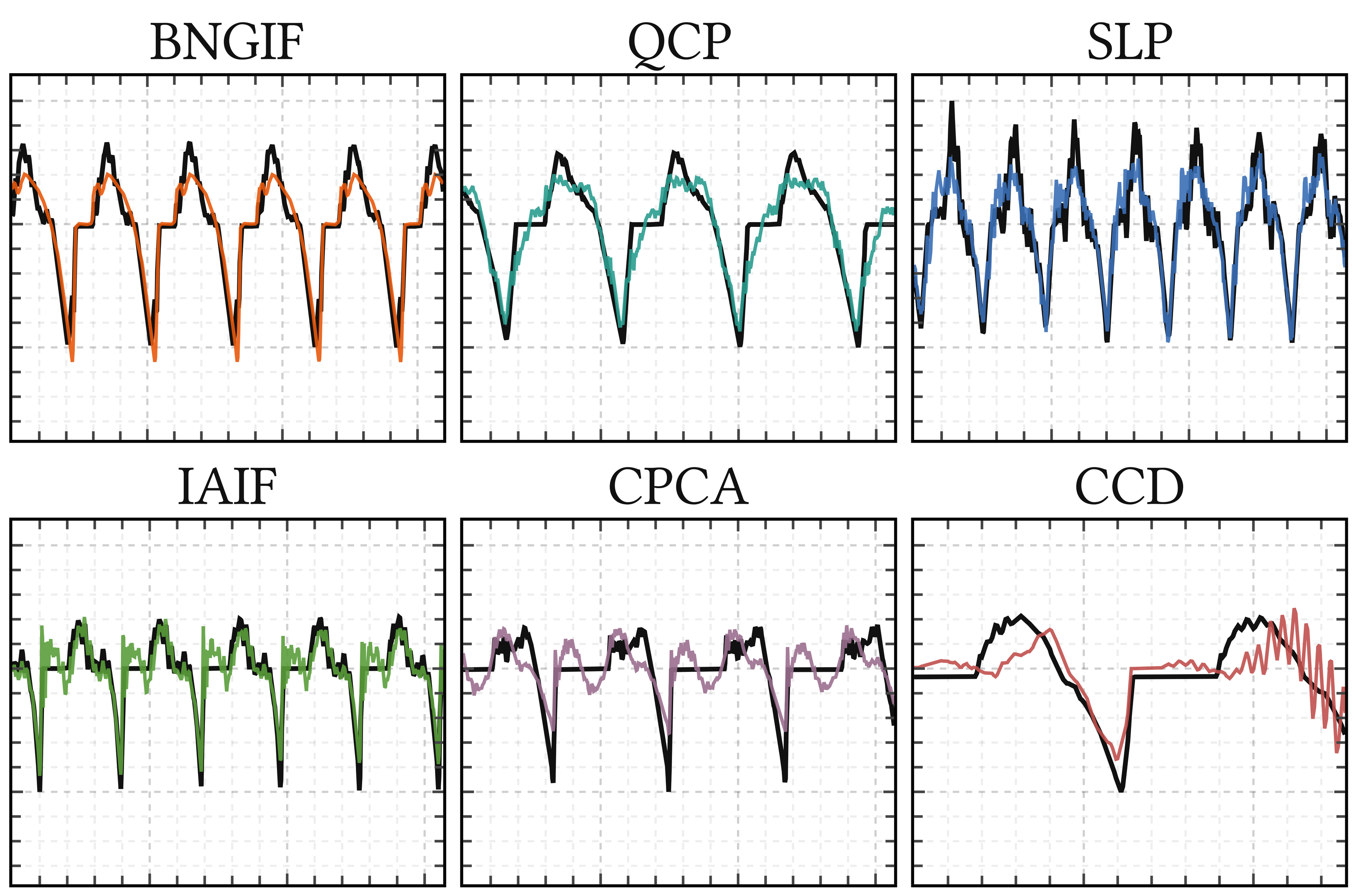
PROBABILISTIC SPEECH DECONVOLUTION IN REAL TIME

METHOD





BENCHMARK



True airflow (black) and median performance of six methods' estimates on EGIFA. Our method = BNGIF; other 5 are current state of the art.

	BNGIF	QCP	SLP	IAIF	CPCA	CCD
BNGIF	**0.935**	<.001	<.001	<.001	<.001	<.001
QCP	1.000	**0.910**	<.001	<.001	<.001	<.001
SLP	1.000	1.000	**0.889**	<.001	<.001	<.001
IAIF	1.000	1.000	1.000	**0.886**	<.001	<.001
CPCA	1.000	1.000	1.000	1.000	**0.759**	0.004
CCD	1.000	1.000	1.000	1.000	0.996	**0.741**

Paired Wilcoxon per utterance. Diagonal: median score on EGIFA. Off-diagonal: p against "row is no better than column". Significant p means row consistently outscores column.

APPLICATIONS

Clinical medicine
Extract high-precision physiological voice biomarkers with calibrated uncertainty.

Phonetics and forensics
Formant estimates 30% more accurate (RMSE) than Praat.

Hearing implants?
Cheap model for detecting speech over noise. On-device decoding?

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interactive figures